

SCIENCE ONE, MATHEMATICS - HOMEWORK #2.1

Due on Friday, January 30th at 11PM.

1. Suppose that f is differentiable with $f'(x) > 0$ for all x and suppose that $f(1) = 0$. Set

$$F(x) = \int_0^x f(t) dt.$$

Determine if each statement is True or False. Justify your answer.

- (a) The function $F(x)$ is continuous.
- (b) The function $F(x)$ is not twice differentiable.
- (c) The value $x = 1$ is a critical point of F .
- (d) The function $F(x)$ takes on a local minimum at $x = 1$.
- (e) $F(1) > 0$.

Based on your answers above, make a rough sketch of the graph of $F(x)$.

2. Recall in class we proved that $\sum_{i=1}^k i = \frac{k(k+1)}{2}$. In this problem, we will construct a proof of

$$S = \sum_{i=1}^k i^2.$$

- (a) Calculate the value of S for $k = 1, 2, 3$, and 4 .
- (b) Calculate the value of $\frac{k(k+1)(2k+1)}{6}$ for $k = 1, 2, 3$, and 4 .
- (c) Show that $\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$ by constructing a proof by induction by:
 - i. showing the equality is true for the base case (i.e., $k = 1$).
 - ii. for each positive integer N , the equality holds for $k = N$ implies the equality for $k = N + 1$.

Be sure to end your proof with an appropriate conclusion statement (i.e., “Therefore, by induction,”).

3. Let $f(x)$ be an integrable function over the interval $[a, b]$. Choose an integer N and let

$$x_k = a + k\Delta x, \quad k = 0, 1, 2, \dots, N \text{ where } \Delta x = \frac{b-a}{N}.$$

The trapezoid rule is an approximation of the definite integral

$$\int_a^b f(x) dx \approx \Delta x \sum_{k=1}^N \frac{f(x_k) + f(x_{k-1})}{2},$$

where $\Delta x = \frac{b-a}{N}$.

A bound on the error is given by the error formula

$$\left| \int_a^b f(x) dx - \frac{b-a}{N} \sum_{k=1}^N \frac{f(x_k) + f(x_{k-1})}{2} \right| \leq \frac{(b-a)^3}{12N^2} K_2,$$

where K_2 is any number such that $|f''(x)| \leq K_2$ for all $x \in [a, b]$.

Consider the Fresnel integral

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2) dx.$$

- (a) Write out and calculate the Left and Right Riemann sums for $N = 4$.
 - (b) Write out and calculate the Trapezoid rule for $N = 4$.
 - (c) Find an appropriate value of K_2 for the Fresnel integral. Be sure to justify your answer.
 - (d) Find the value of (the smallest) N which guarantees the trapezoid rule approximates the Fresnel Integral with errors less than 10^{-1} and 10^{-2} . Be sure to justify your answers.
4. In class, we proved Part 2 of the Fundamental Theorem of Calculus, which states that if f is a continuous function on the interval $[a, b]$ and F is any antiderivative of f , then

$$F(b) - F(a) = \int_a^b f(x) dx.$$

Here, you will construct another proof of Part 2 of the Theorem.

- (a) Divide the interval $[a, b]$ into n subintervals with endpoints $a = x_0 < x_1 < x_2 < \dots < x_n = b$. Show that

$$F(b) - F(a) = \sum_{i=1}^n (F(x_i) - F(x_{i-1})).$$

- (b) Now suppose that F is any antiderivative of f . Show that there exists a number c_i in each interval $[x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1}).$$

- (c) Now build an appropriate Riemann sum for f on $[a, b]$ and show that

$$F(b) - F(a) = \int_a^b f(x) dx.$$